EduAssist AI — LLMOps Assignment 2

API Cost, Latency, and Prompt Version Management

Course: MLOps | Individual experiential learning assignment

Objective and design. This assignment operationalizes an academic question-answering assistant by measuring request latency, token consumption, API cost, and prompt compliance. Three prompt versions are compared on the same five undergraduate machine-learning questions: v1 is a basic instruction, v2 adds an undergraduate-friendly role and style, and v3 requires a strict JSON object containing an answer, confidence label, and key points. The design uses a common question set to make the prompt comparison controlled and reproducible.

Task 1 — latency. The notebook records question ID, prompt version, model, UTC start and end timestamps, latency, and response status. It calculates mean, median, minimum, maximum, and standard deviation for each prompt version. The included CSV is a simulated preview, not a live API benchmark; latency is deliberately marked as zero in the no-key path. A final submission must run the notebook with an API key to obtain real latency observations and identify the slowest request. Potential causes include network variation, model service load, prompt length, and generated output length. Five requests per version are adequate for a classroom demonstration but not for a production latency SLA.

Task 2 — token and cost economics. The notebook extracts input and output token usage from each API response. Input cost equals input tokens divided by one million and multiplied by the input price per million; output cost is calculated in the same way. The request cost is their sum, and projected monthly cost is average request cost multiplied by 10,000. The default reference rates are USD 0.40 per million input tokens and USD 1.60 per million output tokens for GPT-4.1 mini, checked on 31 October 2026. Prices and model availability can change, so the notebook exposes both rates as environment-variable overrides. The projection excludes retries, taxes, and other optional service charges.

Task 3 — prompt versioning and MLflow. The experiment is named *eduassist_llmops_assignment*. For each prompt, the notebook logs average latency, average input tokens, average output tokens, average request cost, and the JSON validity metric. The v3 validator checks that the response parses as JSON and that the required fields have the expected types and allowed confidence value. v1 and v2 are plain-text prompts, so JSON validity is not applicable to them. The notebook attempts to register all three prompts with MLflow's GenAI Prompt Registry and assign the *production* alias to v3. Registry support depends on the installed MLflow release; the run output must be checked before claiming registration or promotion succeeded.

Recommendation and interpretation. Prompt v3 is the recommended production candidate when the consuming application requires machine-readable output, because it explicitly defines the JSON contract and can be automatically validated. Promotion should remain conditional on a live run showing an acceptable JSON pass rate, factual answer quality, latency, and cost. v1 may be cheaper because it is shorter, while v2 may be easier for students to understand; neither guarantees the required structured response. A practical cost optimization is to remove redundant prompt text, cap response length, cache stable context when supported, and route simple questions to a smaller model only after validating answer quality.

Limitations and conclusion. JSON validity does not prove factual correctness, and a five-question sample is too small to establish robust quality or reliability. The preview data bundled with this package is clearly labelled and must be replaced with live outputs for final assessment. Overall, the experiment demonstrates the LLMOps principle that prompts are production artifacts: they should be versioned, evaluated, measured for cost and latency, and promoted through a traceable process rather than edited informally.

References

OpenAI API model documentation: https://developers.openai.com/api/docs/models/gpt-4.1-mini
OpenAI API pricing: https://developers.openai.com/api/docs/pricing
MLflow documentation: https://mlflow.org/docs/latest/